

PROJECT REPORT

Java Library Management System

Course: Programming in Java

Academic Year: 2026-27

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1. Introduction

The Java Library Management System is a command-line application developed to demonstrate the major Java concepts included in the supplied course syllabus. It models library operations such as registering books and members, issuing and returning books, saving records and storing circulation transactions in a database.

2. Problem Statement

A library needs a structured method to maintain book and member information and track circulation. The objective is to build a simple Java application that performs these operations while demonstrating core programming and database concepts.

3. Objectives

Apply Java fundamentals; use classes, objects and constructors; demonstrate encapsulation, inheritance and polymorphism; implement interfaces and custom exceptions; use collections and file I/O; demonstrate multithreading; connect Java with SQLite using JDBC; and provide command-line execution and testing.

4. Technologies Used

Java 17+, Maven, SQLite, JDBC and JUnit 5.

5. Syllabus Mapping

Unit 1: Java fundamentals, variables, data types, operators, input/output, conditions and loops. Unit 2: classes, objects, constructors, access modifiers, this, inheritance, overriding, abstract classes, interfaces, polymorphism and encapsulation. Unit 3: exception handling and multithreading. Unit 4: ArrayList, collections and I/O streams. Unit 5: JDBC connection, SQL, prepared statements and result sets.

6. Methodology

The application initializes the SQLite database, loads saved records, displays a command-line menu, accepts user input, validates operations, updates application state, records circulation transactions through JDBC, and saves persistent records.

7. Testing

JUnit tests verify adding books and members, issuing and returning a book, and preventing an unavailable book from being issued again. Tests can be run using mvn test.

8. Expected Results

The application should add and list records, issue and return books, reject invalid operations, persist records, store JDBC transactions, display database records and execute the statistics task.

9. Limitations

This is an academic prototype and does not include authentication, a graphical interface, network-based multi-user operation or production-grade database administration.

10. Future Scope

JavaFX GUI, authentication and roles, fine calculation, search/filtering, MySQL/PostgreSQL support, REST API and multi-user deployment.

11. Conclusion

The project combines the major topics of the supplied Programming in Java syllabus into one executable application, demonstrating Java fundamentals, OOP, exception handling, collections, I/O streams, multithreading and JDBC.

12. References

Oracle Java Documentation; JDBC API Documentation; JUnit 5 User Guide; SQLite JDBC documentation; course material for Programming in Java.